

GEOSCIENCE EDUCATION IN ACTION: A REVIEW OF AN INTERNATIONAL COLLABORATION BETWEEN HOFSTRA UNIVERSITY (USA) AND EUROPEAN UNIVERSITIES (EU)

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Abstract

Greece is an excellent natural lab for attracting students in geoscience. Students investigate how Earth works at the strategic tectonic boundary where the African-Eurasian tectonic plate interaction triggers a wide range of natural hazards likewise in other global locations as well. Earthquakes, volcanic eruptions, and other surface processes are imminent, and the active tectonic and the volcanic Hellenic region requires continuous monitoring, an advanced hazard evaluation, and active dissemination of knowledge to society. Students learn how to implement a hazard evaluation and management that reinforces the societal resilience and facilitates mitigation against geological phenomena that threaten the infrastructure.

An annual international collaboration takes place by the outcome of students and faculty who join forces to learn and promote geoscience utilizing advanced technology and modern techniques. We describe the outcome of a collaboration between the Hofstra University in New York (USA), and the Hellenic institutes, the National and Kapodistrian University of Athens (NKUA), the National and Technical University of Athens (NTUA) to participate in monitoring active volcanoes and tectonic boundaries.

Expeditions take place at our current operating stations at the recent HVA (Hellenic Volcanic Arc) and at the HFA (Hellenic Fore Arc). At the HVA, stations monitor the volcanoes at Sousaki, Methana, Milos, Santorini, and Nisyros. At the HFA, multiple stations are located in Kythera, Antikythera, and Molaoi, while more stations are monitoring at the Gulf of Korinthos and Saronikos in Desfina (Amfissa), Velo (Korinthos), and Makriloggos (Methana).

We provide to the local and international students the opportunity to conduct research and present their outcome by using advanced technology such as GPS, 3D reconstruction, mapping, digital elevation models, geochemical analysis, drone thermal-scanning, temperature monitoring of active fumaroles, understanding the local weather stations affecting volcanic measurements, and data processing in a big data lab with an on-campus or long-distance accessibility.



Figure 1: Launching an autonomous drone for automatic thermal night-scanning inside the Stefanos crater on Nisyros Volcano with students from Hofstra University and National and Kapodistrian University of Athens.. From left to right: Prof. Bret Bennington (HU), Prof. Konstantinos Kyriakopoulos (NKUA), Tom Plitnick (HU), Daniella Chernoff (HU), Morgan Jones (HU), Konstantinos Hantzis (NTUA), Prof. Antonios Markellos (HU).

Introduction

- ✓ The Hellenic volcanic arc and forearc are an excellent natural lab for students applying Geoscience methods.
- ✓ A sustainable future for geo-education is based on the successful collaboration between educational institutions, local authorities and government, and local people.
- ✓ A resilient society against natural hazards is founded on knowledge dissemination and adequate monitoring of natural phenomena including but not limited to volcanic activity.
- ✓ Here we show a quick look of our successful collaboration.

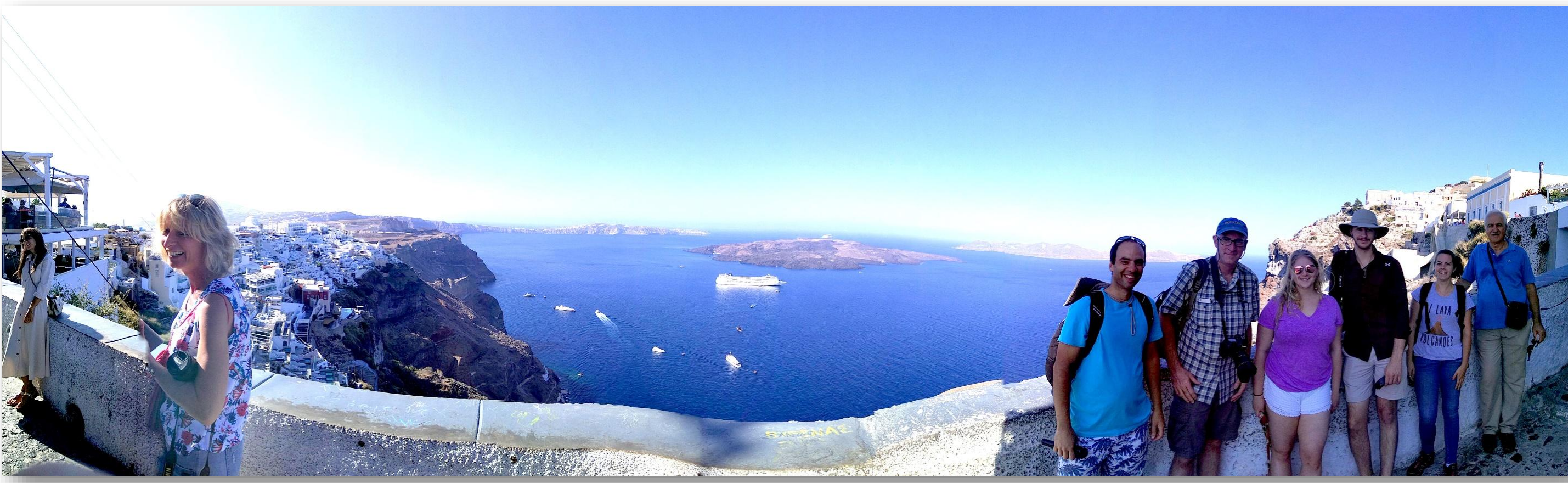
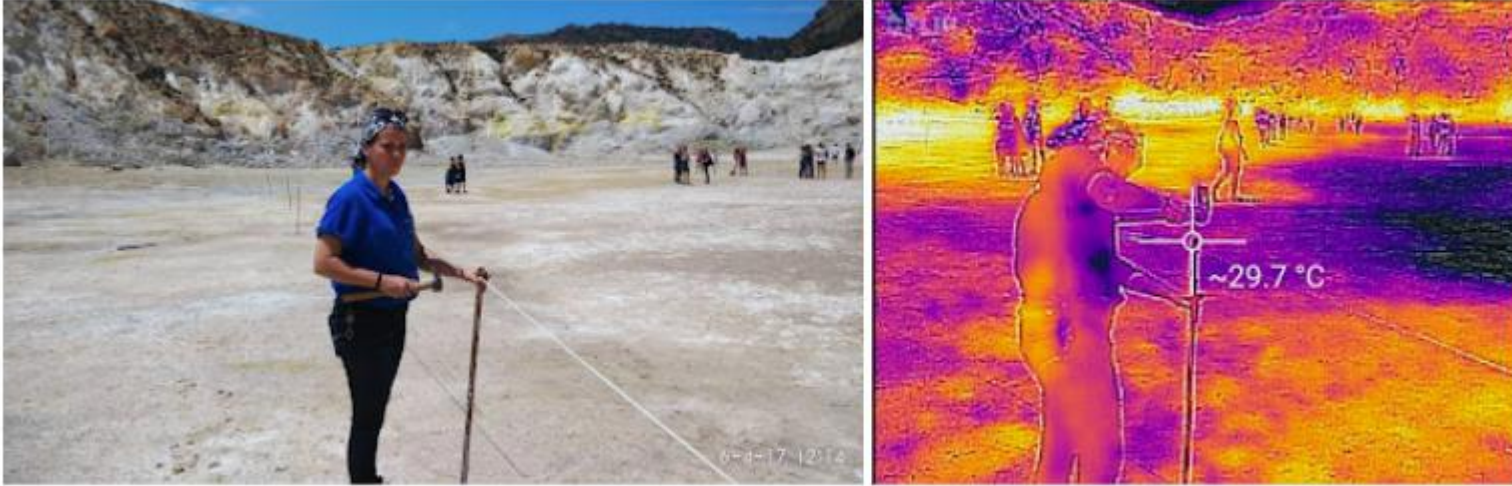


Figure 2: At the center is the monitoring station at Santorini Caldera. International research team visiting the Greek volcanoes, Santorini and Nisyros islands. (from right to left) Prof. K. Kyriakopoulos from National University of Athens, and Morgan Jones graduate student, Tom Plitnick and Daniella Chernoff undergraduate students, and Prof. Bret Bennington (chair) and Prof. Antonios Marsellos from the department of Geology, Environment and Sustainability at Hofstra University.

Methods

- Providing students:
- Experience at the field and lab. Funding using internal (Hofstra University), external (National Science Foundation) grants, and local people's support to ensure application of high-technological and advanced geological methods at the field for USA and EU students.
- Geographical Information Systems applications on geochemical data at active volcanoes.
- In-situ Portable X-ray Fluorescence sampling and geochemical mapping
- Evaluation of a sustainable tourism by using statistical analysis of demographic questions at residential areas at the active volcanic regions.
- Students learn how to use SPSS, R, ArcMap, GPS, and Drone software for geodata analysis and digital elevation modeling.
- Mapping active faults and extensional structures on active volcanic sites.
- Students learn how to fly drones for thermal scanning and studying active fumaroles.



- Students learn how to geochemical sample and map of a volcanic area using thermal scanners at the night.



Results

Mapping active extensional structures on active volcanic sites.

GPS and Temperature data collection by Faith Renner (student)

Nick Platak and Faith Renner repairing the weather station at the Sousaki Volcano

Hofstra USA-students and EU-students meet at the active Volcano for international research

NSF-Geoteam in action installing a new thermoprobe on the Santorini Volcano

Greek and American professors with students repairing Santorini station.

Testing and retrieving data from the monitoring stations of active fumaroles

Conclusion

- ✓ The utilized methodology seem to be very effective as students gain a significant experience from the field because they feel part of an international team.
- ✓ Students share and implement their own ideas as being for a first-time scientists-in-action collecting their data using telemetry.



Students visiting local museums and see fundamental geo-knowledge at the visited active volcanoes.